Problem 5: Fairness Transfer across States and Years

ICT 5143: AFGML Capstone

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Research question and decision context

- A classifier made fair in one state or year is deployed in another.
- $Y = 1$: income $>$ \$50k (ACSIncome). $D = 1$: high-income prediction (a benchmark, not merit).
- **Question:** with only 200 or 1,000 target labels, does updating ≤ 2 Bayesian-network factors keep equalized odds better than global adaptation?
- **Falsified if** the Holm-adjusted paired bootstrap shows no gain, or balanced accuracy drops by more than 0.02.

Closest work and proposed contribution

- Fairness breaks under distribution shift (Ding et al. 2021; Shao et al. 2024).
- Covariate reweighting assumes $P(Y \mid X)$ stays fixed (Singh et al. 2021). Reproduced as **IW-EO**.
- Closest idea: Schrouff et al. (2022) *diagnose* transfer failures with a graph, but do not repair them.
- **BN-LFA (ours):** update only the ≤ 2 shifted factors that matter for the decision, then refit the fair thresholds.

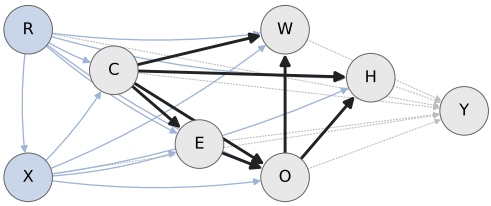
Data and evaluation population

- ACS PUMS 1-Year, CA, TX, NY $\times$ 2018, 2019: six domains (Folktables check passed).
- 18 groups: recorded race $\times$ sex. Split by household: 60 / 20 / 20 %.
- Target labels come only from the train + validation pool; the test split only evaluates.

	CA18	TX18	NY18	CA19	TX19	NY19
persons	195,665	135,924	103,021	197,193	138,293	104,330
groups $\geq$100 pos./neg. (test)	10	8	7	10	8	8

- Small groups (American Indian, Alaska Native, Pacific Islander) are reported, but only as exploratory.

Graphical model and inference



Learned in all six domains. Blue: required R, X edges.

- All 8 variables observed; the graph is associational.
- Composition: $P(v \mid R, X, \mathrm{pa}'(v))$, Dirichlet-smoothed.
- Outcome: log-linear $P(Y \mid \text{all features})$.
- Query $P(G, \mathrm{pa}(Y), Y)$ gives each group's TPR and FPR.
- VE $=$ clique tree $=$ enumeration $=$ pgmpy (error 1.3×10^{-15}).

Fairness criterion and proposed method

- **Fairness:** $\Delta_{\mathrm{EOdds}} = \max(\text{TPR gap}, \text{FPR gap})$ across groups. Errors should not depend on the group, given the true label.
- **Utility:** balanced accuracy; at most 0.02 loss allowed.
- **BN-LFA** for each factor: Bayes factor of a shift $\times$ its exact effect on the fair rule, then update the top ≤ 2 factors.
- Refit thresholds on source data weighted by $w = \prod_{v \in F} \hat{\theta}_t / \theta_s$.
- Uses race and sex at prediction time (features and group thresholds).

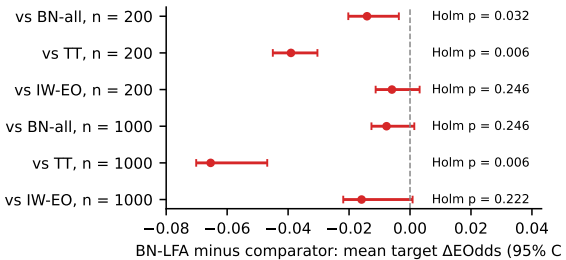
Experimental protocol

- **Baselines:** LR, GBT, reweighing, EO-S (keep source rule), TT (target thresholds), IW-EO, retraining, BN-S, BN-all.
- Every method gets the same n target persons. Test labels are never used for fitting.
- 9 transfers $\times$ 6 budgets $\times$ 5 seeds.
- Paired household bootstrap ($B = 2000$); Holm over 6 confirmatory tests.
- Plus a synthetic simulation, ablations and robustness tests.

Main result and uncertainty

Method, $n = 1000$	BA	Δ_{EOdds}
EO-S (source rule)	0.771	0.150
TT	0.772	0.205
IW-EO (closest)	0.769	0.156
BN-all	0.770	0.147
BN-LFA	0.770	**0.140**

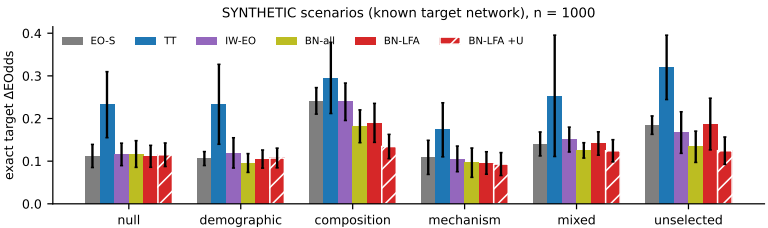
Mean of 9 transfers $\times$ 5 seeds; lower Δ_{EOdds} is fairer.



Answer: partially supported. BN-LFA is fairer than TT (both budgets) and BN-all ($n = 200$), but not than IW-EO or the unchanged source rule.

Ablation, robustness and failure case

- **Ablation** ($n = 1000$): random factors 0.152 and all factors 0.147, vs BN-LFA 0.140.
- **Mechanism:** the diagnostic predicts the fairness change (Spearman 0.88).
- **Failure:** three shifted factors but only two allowed: no better than EO-S (0.187 vs 0.184).



SYNTHETIC: with unlabelled data, BN-LFA fixes a group-specific shift that IW-EO cannot (0.134 vs 0.220).

Conclusion and reproducibility

- **Conclusion:** with few labels, adapting ≤ 2 factors beats refitting thresholds, but gives no reliable gain over reweighting or no adaptation.
- **Limits:** associational graph; 3 states, 2 years; small groups; the \$50k label; race and sex used at decision time.
- **Reproduce:** *Run all* in `problem5_fairness_transfer_bn.ipynb` (Google Colab, 13 min).
- **Aslam Mugutkhan:** [*contribution and verification*].
- **Rao Ashlesha Padmananda:** [*contribution and verification*].

Backup: mandatory evidence and references

Requirement (Problem 5)	Evidence in `results/`
E1–E3: within-domain, transfers, budgets	`exp_transfer_summary.csv`, Figs. 2–3
E4: composition vs mechanism simulation	`sim_all_runs.csv`, Fig. 7
E5: method comparison	`confirmatory_tests_holm.csv`
Ablations; robustness tests	`ablation_d_eodds.csv`, `robustness_summary.csv`

Full matrix: `docs/requirements_matrix.md`. Key references: Ding et al. 2021; Shao et al. 2024; Hardt et al. 2016; Schrouff et al. 2022; Singh et al. 2021; Pearl & Bareinboim 2011 (16 papers in `docs/related_work.md`).

Backup: inference and statistical validation

- CPTs normalised; factor algebra, evidence and missing values checked (notebook self-checks).
- 20 random queries: all exact methods agree to 1.3×10^{-15}. Elimination order changes cost only.
- Every local Markov statement is a d-separation; largest residual dependence 0.047 bits.
- Model-based group TPRs within sampling noise for 89 % of groups.
- Bootstrap resamples households; the same resamples for all methods (paired).

Backup: demonstration and individual defence

- Run notebook section 2.12 (a few seconds).
- $P(Y{=}1 \mid \text{Black, female, graduate}) = 0.6636$ by VE, clique tree and enumeration.
- CA18→TX18 with 200 labels: BN-LFA updates E and R; Δ_{EOdds} 0.143 vs 0.156 (EO-S), 0.237 (TT).
- Why R barely matters: equalized odds conditions on the group, so race mix alone does not move error rates.
- Saved trace: `demo/demo_trace.txt`.